24/03/26

Day - 8

1)

Average of 5 numbers = 20

Average = $\frac{\text{Total}}{n}$

total = Avg × n

= 20 × 5

$\boxed{\text{total} = 100}$

2)

Avg of 10 numbers = 15

Total of 10 numbers = 10 × 15 = 150

(n−1)

Avg = 14 for 9 number

= 9 × 14

Total = 126

Removed number = 150 − 126 = 24

$\boxed{\text{Answer} = 24}$

3)

Milk : water = 3 : 2

$3x : 2x = \frac{3x}{2x}$

⇒ Add 10L water = 3 : 4

$\frac{3x}{2x+10} = \frac{3}{4}$

$3x(4) = 3(2x+10)$

$12x = 6x + 30$

$12x - 6x = 30$

$6x = 30$

$x = 30/6$

$\boxed{x = 5}$

Milk = $3x = 3(5) = 15L$

water = $2x = 2(5) = 10L$

total = $15 + 10 = 25L$

4)

$$Avg = \frac{First + Last}{2}$$

$$= \frac{1+20}{2} = \frac{21}{2} = 10.5$$

$\boxed{\text{Answer} = 10.5}$

5)

Avg of 5 students = 18

Total = $5 \times 18 = 90$

$(n+1)$

$19 \times 6 = 114$

$114 - 90 = 24$

$\boxed{\text{Answer} = 24}$

Block 2.

1) They help organize and store data efficiently

2) Searching, inserting, deleting

3) By choosing the right structure, operations become faster and more efficient

Grammar correction:

1) He doesn't understand data structures

2) She has completed the assignment

3) They were working on the project

Block 3:

My favorite programming language "Java".

I like java because it is simple, secure, and widely used

One main advantage is platform independence which means Java programs can run on any system

Java is also objected oriented which helps in writing clean and reusable code

I ~~solved~~ problems like sorting and array based programs,

I conclusion

which helped me understand
data structures later.